

Week 3 Day 1

How a single web search actually work?

→ Step 1: You Enter the URL
(Uniform Resource locator)

https://google.com:80/product?id=10#review

↓ Protocol ↓ Domain ↓ Port ↓ Path ↓ Query

Fragment

Step 2: It check Local browser cache does it have the required ~~data~~ a IP for the server.

Step 3: DNS (Domain Name System) Lookup.

It is like a phonebook but for the internet it keep records of Domain names to there assigned IP. ARecord → IPV4, AAAA → IPV6, CNAME → Alias.

Step 4: Establishing connection to the server

TCP → TLC → HTTP

TCP (Transmission Control Protocol) it ensure creates a connection between applications across a network.



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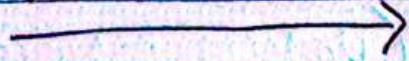
TCP three way handshake

Browser

Server

SYN (synchronize seq num)

SEQ = 5000



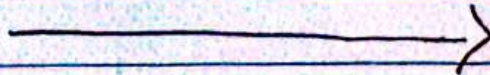
SYN



SEQ = ~~5000~~ 9000

ACK = 5001

ACK = 9001



TCP not only create a connection it

- Sequence numbers
- Acknowledgment
- Reliable delivery
- Retransmission
- Ordering
- Duplicate handling
- Flow control

Segments

TCP divides data into segments. Each segment have it's own SEQ number / ID to identify each segment and keep track of order of data ~~segment~~ received.



TLS (Transport Layer Security)

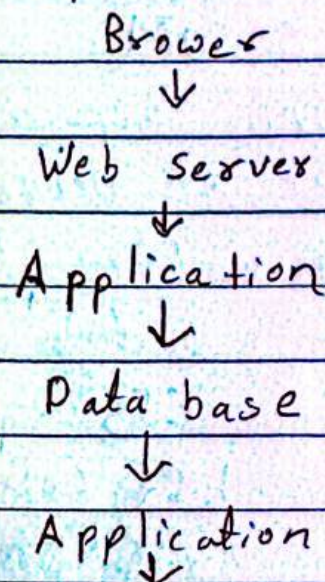
It helps to create a secure connection between client and server.

HTTP Request, Processes and Response

1: The browser sends an HTTP req.

~~get~~ GET /products

2: Server processes the request



HTML response

3: Browser Receives HTML.

The browser receives bytes (01001101 ...)

Then HTML Parser comes in action

Note: Browser decodes those bytes into characters using encoding like (UTF-8).

Step 5: HTML Parser & CSS Parser

1: HTML Parser

It reads HTML code then converts it into a structured format usually a DOM (Document Object Model) tree.

2: CSS Parser

It takes CSS and converts it into CSSOM (CSS Object Model).

Step 6: DOM & CSSOM

DOM & CSSOM represent the HTML document as a tree of elements and CSS rules and style as a tree like structure and combine them creating a Render Tree

HTML

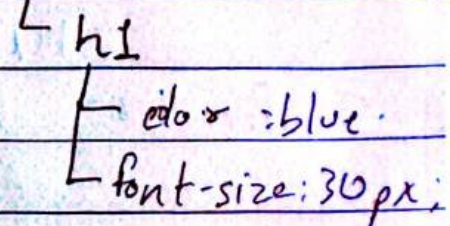
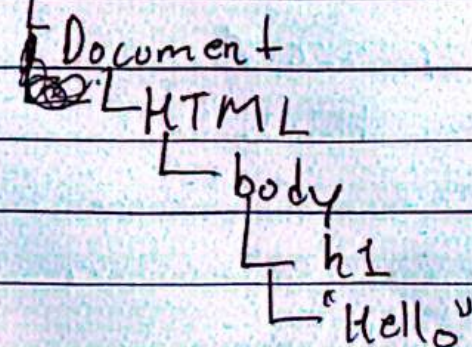
CSS

<h1>Hello</h1>

h1 { color: blue; font-size: 30px; }

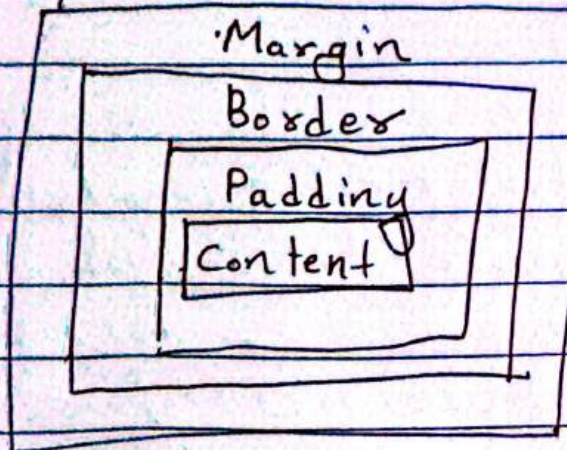
DOM

CSSOM



Step 7: Layout Box Model

Box model describe ~~what~~ how the browser should calculate the size and space taken by an element.



Step 8: Paint, Composite, Page visible

After layout it known, now browser draws the visual parts of the pages, such as Text, colors, Background, Borders, shadow, images, etc. The browser may divides the pages into different layers. The compositor then taken those painted layers and put them together in the correct order. and produce the final frame. After all that we'll get a final visible page which is shown to the user.



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Client vs Server

The client (browser) displays and interacts with the page. The server processes requests, runs backend logic, access databases, and sends responses.

HTML (Hyper text markup language)

It helps to create the structure of a web page. HTML consists of too many tags such as `<h1>`, `<div>`, `<p>`, `<span>`, `<input>`, `<form>`, `<button>`.

CSS (Cascading Style sheets)

It is used to style and design a web page, such as colors, fonts, and layouts. CSS have multiple selector id selector (`#`), class selector (`.`), Element selector, combinator selectors, Attribute selector.

Java Script

It is a programming language used to make web page interactive and dynamic web.

Variables are the containers used to store data. ~~let~~ let, var, const

~~if Var~~ - Functions: A reusable block of code that perform a specific task.

Event: Actions that happen on a webpage, such as clicking, typing, or submitting.

DOM (Document Object Model). A representation of the HTML page that JavaScript can read and modify

Browser Dev Tools: Built in browser tools ^{used} ~~and~~ to inspect, test, and debug websites.

Elements Tab, Console Tab, Network Tab, Application storage.

401 - Unauthorized

404 - Not found

400 - Bad Request

403 - Forbidden

500 - Internal Server Error

~~200~~ 2xx → Success

3xx → Redirection

4xx → Client / request problem

5xx → Server problem.